Single-Peak Crow–Kimura Dynamics: Complete Finite-Genome Spectrum and Exact Projective Gap

Route-A source-local certificate HCS-C336

3 September 2026

Abstract

For the continuous-time mutation–selection equation on binary strings of length L, with symmetric point mutation and one master-sequence fitness spike, we identify the nonlinear simplex flow exactly with a projectivized positive linear semigroup. Walsh decomposition shows that each Hamming eigenvalue retains an explicit codimension-one multiplicity, while all remaining eigenvalues are the roots of one degree-$(L+1)$ secular polynomial. The secular roots are simple and strictly interlace the mutation poles. The two largest roots therefore give the exact projective spectral gap, with a sharp generic convergence law.

1 Frozen mutation–selection owner

Let $\mathcal{H}_L = \mathbb{R}^{\{0,1\}^L}$, let F_i flip bit i, and let e_0 denote the all-zero genotype. For $L \geq 1$ and $U, s > 0$ define

$$M_L = \frac{U}{L}\sum_{i=1}^{L}(F_i - I), \qquad A_L = M_L + s e_0 e_0^\mathsf{T}. \tag{1}$$

Both matrices are real symmetric and $1^\mathsf{T} M_L = 0$. On the probability simplex the Crow–Kimura equation in this normalization is

$$\dot{p} = A_L p - (1^\mathsf{T} A_L p)p = A_L p - s p_0 p. \tag{2}$$

Theorem 1 (Exact projectivization and global limit)**.** *For every probability vector p^{in},*

$$p(t) = \frac{e^{tA_L}p^{\text{in}}}{1^\mathsf{T} e^{tA_L}p^{\text{in}}} \tag{3}$$

is the unique solution of (2). The denominator is positive, the simplex is preserved, and $p(t)$ converges to the uniquely normalized positive Perron eigenvector of A_L.

Proof. Write $q(t) = e^{tA_L}p^{\text{in}}$ and $Z(t) = 1^\mathsf{T} q(t)$. Positivity of the irreducible hypercube semigroup gives $q(t) > 0$ for $t > 0$, hence $Z(t) > 0$. Moreover $Z' = 1^\mathsf{T} A_L q = s q_0$. Differentiating q/Z gives exactly (2). The same calculation shows that its entries sum to one. Since A_L is irreducible Metzler and symmetric, its largest eigenvalue is simple and has a strictly positive eigenvector. The Perron coefficient of every nonzero nonnegative initial vector is positive; dividing the spectral expansion by $Z(t)$ proves the limit. ☐

The phrase *projective semigroup owner* records the theorem content of the original manuscript round.

2 Complete finite-genome spectrum

For $S \subseteq \{1, \ldots, L\}$ let

$$\phi_S(x) = 2^{-L/2}(-1)^{\sum_{i\in S} x_i}.$$

These Walsh characters form an orthonormal basis and

$$M_L \phi_S = d_{|S|}\phi_S, \qquad d_k = -\frac{2Uk}{L}, \qquad \langle e_0, \phi_S\rangle = 2^{-L/2}. \tag{4}$$

Put $m_k = L!/[k!(L-k)!]$, $w_k = m_k/2^L$.

Theorem 2 (Full multiplicity and secular factorization)**.** *The complete characteristic polynomial of A_L is*

$$\det(\lambda I - A_L) = q_L(\lambda) \prod_{k=0}^{L} (\lambda - d_k)^{m_k - 1}, \tag{5}$$

$$q_L(\lambda) = \prod_{k=0}^{L} (\lambda - d_k) - s \sum_{k=0}^{L} w_k \prod_{j \neq k} (\lambda - d_j). \tag{6}$$

Thus d_k is retained with multiplicity $m_k - 1$ and the remaining $L + 1$ eigenvalues are precisely the roots, away from the poles, of

$$1 = s \sum_{k=0}^{L} \frac{w_k}{\lambda - d_k}. \tag{7}$$

The multiplicities in (5) sum to 2^L.

Proof. Let $E_k = \operatorname{span}\{\phi_S : |S| = k\}$. The subspace of E_k whose Walsh coefficients sum to zero is orthogonal to e_0 and has dimension $m_k - 1$; A_L acts there by d_k. Its orthogonal complement over all k has basis

$$u_k = m_k^{-1/2} \sum_{|S|=k} \phi_S, \qquad \langle e_0, u_k \rangle = \sqrt{w_k}.$$

The restriction of A_L is $D + svv^{\mathsf{T}}$, where $D = \operatorname{diag}(d_0, \dots, d_L)$ and $v_k = \sqrt{w_k}$. The matrix determinant lemma gives (6); adjoining the retained orthogonal spaces gives (5). ☐

3 Strict interlacing and projective rate

Order the mutation poles as $d_L < \cdots < d_1 < d_0 = 0$ and set $R(\lambda) = s \sum_k w_k/(\lambda - d_k)$. On every pole-free interval,

$$R'(\lambda) = -s \sum_{k=0}^{L} \frac{w_k}{(\lambda - d_k)^2} < 0. \tag{8}$$

At the left and right ends of each internal gap, R has limits $+\infty$ and $-\infty$, respectively. Above d_0, it decreases from $+\infty$ to 0; below d_L it is negative.

Theorem 3 (Interlacing and exact projective gap)**.** *The $L + 1$ secular roots $\rho_0 > \rho_1 > \cdots > \rho_L$ are simple and obey*

$$\rho_0 > d_0 = 0 > \rho_1 > d_1 > \rho_2 > \cdots > d_{L-1} > \rho_L > d_L. \tag{9}$$

In particular, the two largest eigenvalues of the full 2^L-dimensional matrix are ρ_0 and ρ_1. With

$$\Delta_L(U, s) = \rho_0 - \rho_1, \tag{10}$$

every simplex orbit satisfies $p(t) - p_\infty = O(e^{-\Delta_L t})$. This exponent is attained whenever the coefficient of the ρ_1 eigenvector in the projective expansion is nonzero.

Proof. Equation (8) and the one-sided pole limits give exactly one root in every internal gap, one above d_0, and none below d_L. The retained eigenvalue nearest the top is $d_1 < \rho_1$, so ρ_1 is the true second eigenvalue. Divide the real-symmetric spectral expansion in Theorem 1 by its Perron term. All remaining exponents are at most $\rho_1 - \rho_0$; a nonzero ρ_1 coefficient gives the stated leading term. ☐

The phrase *strict secular interlacing* records the substantive Round-1 addition.